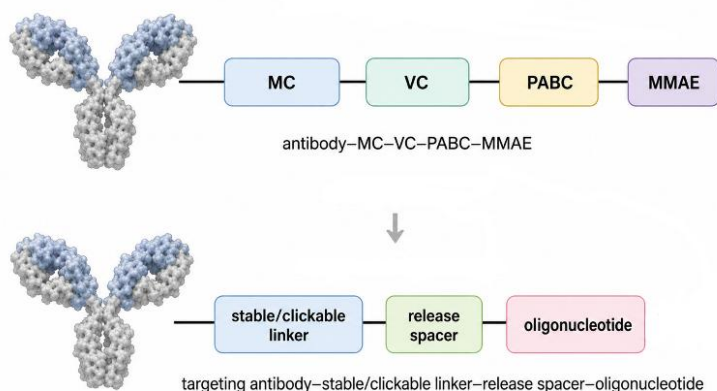


maleimide group reacts with reduced cysteine thiols on the antibody to form the covalent ADC linkage. The VC segment, valine–citrulline, is a protease-cleavable dipeptide designed to remain stable in circulation but be cleaved after internalization into target cells, particularly in the lysosome. PABC, p-aminobenzyl alcohol/carbamate, acts as a self-immolative spacer that breaks down after VC cleavage to release the free payload. MMAE, monomethyl auristatin E, is the potent antitubulin drug that disrupts microtubule dynamics and kills the CD30-positive cancer cell after release. The uploaded review describes BV/Adcetris as MMAE conjugated to a CD30-targeting monoclonal antibody through a maleimidocaproyl–valine–citrulline–PABC linker.



To develop a new ADO, the same modular logic could be retained, but the small-molecule payload MMAE would be replaced with an oligonucleotide payload, such as an siRNA, ASO, splice-switching oligo or PMO. A practical strategy would be to keep an antibody-binding chemistry, such as cysteine–maleimide conjugation or a more stable next-generation cysteine linker, while replacing the MMAE attachment point with a click-compatible handle. For example, the antibody could be functionalized with an azide, alkyne, DBCO, or tetrazine-bearing linker, and the oligonucleotide could be synthesized with the complementary click handle at its 5' or 3' end.

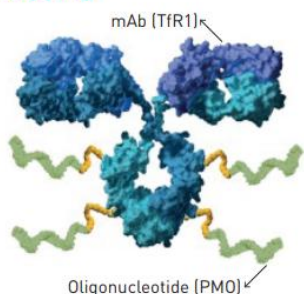
Site-specific attachment

A key technology leap from early ADC/ADO's is the cysteine conjugation where defined cysteine residues are engineered into selected

sites on the antibody heavy or light chain, creating reactive thiol handles that can be selectively coupled to thiol-reactive linker–payloads, thereby enabling site-specific attachment of oligonucleotides with controlled loading, improved conjugate homogeneity, and reduced variability compared with random lysine or native cysteine conjugation (Genentech, 2016).

While conventional antibody conjugation often relies on native lysine residues or reduced interchain cysteines, producing heterogeneous mixtures with variable payload loading and attachment sites, THIOMAB engineering introduces reactive cysteine residues at defined positions within the antibody heavy or light chain; thiol handles for site-specific linker–payload attachment. These engineered cysteines can react with thiol-selective chemistries such as maleimide, iodoacetamide, or pyridyl disulfide linkers, yielding more homogeneous conjugates with tighter control over drug-to-antibody or oligonucleotide-to-antibody ratios, reduced batch variability, and potentially improved plasma stability and therapeutic index. Although the platform was originally developed primarily for antibody–drug conjugates, the same design principles apply directly to antibody–oligonucleotide conjugates, where the placement of ASO, siRNA, or PMO payloads can strongly influence biodistribution, receptor-mediated uptake, intracellular trafficking, and manufacturability (Genentech, 2016).

Figure 1: AOC 1044, an Antibody Oligonucleotide Conjugate Targeting Exon 44 Skipping

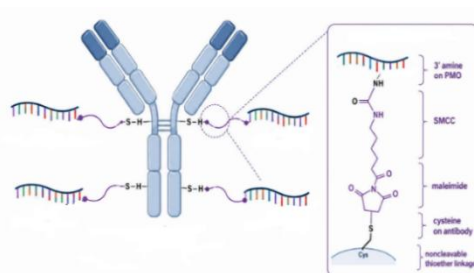


AOC 1044 as a Novel Therapeutic Approach for DMD Patients Amenable to Exon 44 Skipping: EXPLORE44™ Phase 1/2 Healthy Volunteer Data

Mark Stahl, Yiming Zhu, Varun Goel, Laura Leung, Tara Carmack, Yvonne Tami, Trista Hardin, Usue Etxaniz, Philip Kovach, Josiah Herzog, Steve Hughes, Elizabeth J. Ackermann

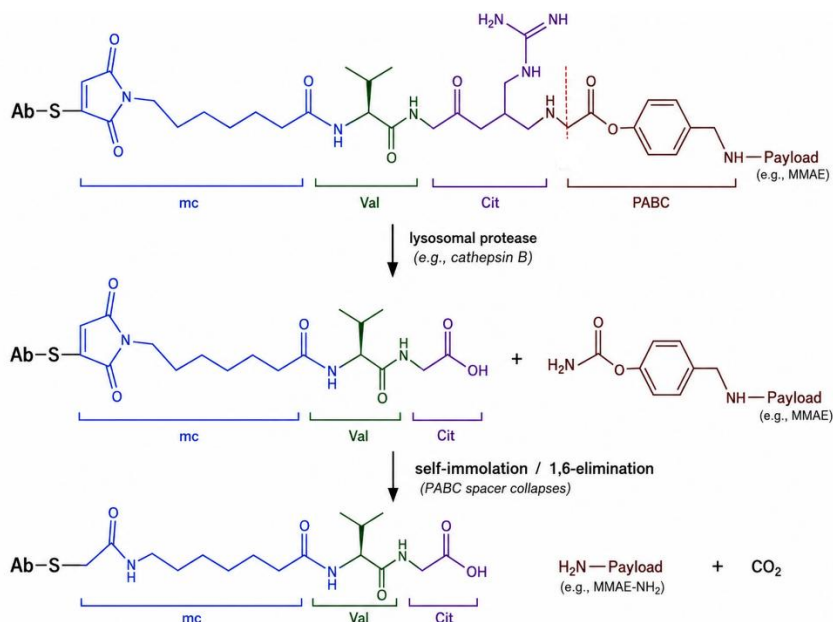
Avidity Biosciences, San Diego, CA, USA

Avidity’s AOC 1044 is a transferrin receptor 1 (TfR1)-targeted antibody–oligonucleotide conjugate developed to deliver an exon 44–skipping phosphorodiamidate morpholino oligomer (PMO) to skeletal muscle. The molecule comprises a humanized anti-TfR1 antibody conjugated to multiple copies of a PMO designed to promote exclusion of exon 44 from dystrophin pre-mRNA. In healthy volunteers, AOC 1044 showed dose-dependent delivery of PMO to skeletal muscle and statistically significant exon 44 skipping after a single dose (Stahl *et al.*, 2024).



Although the precise conjugation chemistry has not been unambiguously disclosed, the most strongly supported publicly available model for AOC 1044 is a noncleavable SMCC/maleimide–thiol architecture. In this scheme, SMCC is coupled to a primary amine at the 3' end of the PMO, generating a maleimide-functionalized PMO that is subsequently reacted with thiols exposed on the anti-human TfR antibody following TCEP-mediated reduction of cysteines. This assignment is consistent with broader AOC linker-design literature, which classifies SMCC as a noncleavable linker and specifically identifies Avidity's industrial AOC linker as SMCC (Li et al., 2025).

Linker technology



The most common clinically approved linker class is the protease-cleavable (maleimidocaproyl)-Val-Cit-PABC linker, usually written as mc-VC-PABC. This linker is used in several MMAE-based “vedotin” ADCs, including brentuximab vedotin, polatuzumab vedotin, enfortumab vedotin, tisotumab vedotin and telisotuzumab vedotin (Jang and Chang, 2025). Its common architecture contains a maleimide-based antibody attachment group, a caproyl spacer, a valine–citrulline dipeptide designed for lysosomal protease cleavage, and a self-immolative PABC spacer that facilitates release of free MMAE. Its clinical success reflects a useful combination of systemic stability and intracellular enzymatic release, although the maleimide–thiol attachment and MMAE payload are also associated with class-specific liabilities such as deconjugation concerns and peripheral neuropathy (Lin et al., 2025; Su et al., 2021).

Approved antibody–drug conjugate (ADC) linkers share a common design objective: they must keep a highly potent payload attached to the antibody during systemic circulation, while enabling payload release after antigen binding, internalization and lysosomal processing in target cells. In this sense, all clinically approved linkers are pharmacological “control elements”, not passive tethers: they influence plasma stability, exposure to free payload, intracellular release, bystander killing and toxicity (Su et al., 2021; Jang and Chang, 2025). Current approved ADCs use both cleavable